(L-25) Generalized Linear Model

$$Y \sim \exp$$

$$f(y) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\} \qquad \mu = E[Y].$$

$$X_1, \ldots, X_d \qquad \eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d$$

$$\eta = g(\mu) \longrightarrow \text{link function}$$

$$\downarrow$$

$$(-\infty, \infty).$$

$$\text{logit} \to \log \frac{t}{1-t}.$$

$Y \sim \text{Poi}(\lambda) \to \mu = \lambda.$

$\hookrightarrow g(t) = \log t.$

eg:

$X_1,$	$\ldots$	$\ldots,$	X_d	Y	Mean
·				y_1	λ_1
·				y_2	λ_2
$\underline{X}_i$	·	$\ldots$	·	·	·
·				·	·
·				y_n	λ_n

Poisson Model

$$\prod_{i=1}^{n} e^{-\lambda_i} \frac{\lambda_i^{y_i}}{y_i!} \quad \curvearrowright \quad \text{only involves } y_i, \lambda_i \text{ currently.}$$

(We need to involve X_{ij})

$$\begin{aligned} E[Y_i] &= \lambda_i = \mu_i \\ \eta_i &= \log \mu_i \\ \eta_i &= \beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di} \\ \lambda_i = e^{\eta_i} &= e^{\beta_0 + \sum_{j=1}^{d} \beta_j X_{ji}} \\ &= e^{\underline{\beta}' \underline{X}_i} \end{aligned}$$

The likelihood function

$$\prod_{i=1}^{n} e^{-e^{\underline{\beta}' \underline{X}_i}} \frac{e^{\underline{\beta}' \underline{X}_i y_i}}{y_i!}$$

$\hookrightarrow$ Maximise w.r.t β. (using Newton Raphson etc.)

Let $\hat{\underline{\beta}}$ be the MLE. For a given $X_0 = (X_{10}, X_{20}, \ldots, X_{d0})$

$$Y \sim \text{Poi}\left(\lambda = \mu = e^{\eta} = e^{\hat{\beta}_0 + \sum \hat{\beta}_j X_{j0}}\right)$$

Now coming back to general exponential family:

$Y \sim$ belongs to one parameter exponential family

$$\hookrightarrow f(y) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

If ϕ is known, it belongs to one-parameter exp. family.

$$\log f = \frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)$$

$$\frac{\partial \log f}{\partial \theta} = \frac{y - b'(\theta)}{a(\phi)}$$

$$E\left[\frac{Y - b'(\theta)}{a(\phi)}\right] = 0$$

$$\Rightarrow \qquad E[Y] = b'(\theta).$$

$$E\left[\frac{\partial \log f(Y)}{\partial \theta}\right] = E\left[\frac{1}{f(Y)}\frac{\partial}{\partial \theta} f(Y)\right]$$

$$= \int \frac{\partial}{\partial \theta} f(y)\, dy$$

$$\begin{pmatrix}\text{exchange of}\\ \text{integral under}\\ \text{regularity}\\ \text{conditions}\end{pmatrix} = \frac{\partial}{\partial \theta}\int f(y)\, dy = 0.$$

$$\frac{\partial^2 \log f(y)}{\partial \theta^2} = \frac{-b''(\theta)}{a(\phi)}$$

$$\left(\frac{\partial}{\partial \theta}\log f(y)\right)^2 = \left[\frac{y - b'(\theta)}{a(\phi)}\right]^2$$

$$E\left[\left(\frac{Y - b'(\theta)}{a(\phi)}\right)^2\right] = \frac{b''(\theta)}{a(\phi)}.$$

$$\Rightarrow \quad E\left[\frac{(Y - E[Y])^2}{a^2(\phi)}\right] = \frac{b''(\theta)}{a(\phi)}$$

$$\Rightarrow \qquad \mathrm{Var}(Y) = a(\phi)b''(\theta).$$

HW Express

i) Ber(p)

ii) Bin(n, p)

iii) Poi(λ)

iv) $N(\mu, \sigma^2)$.

in the form $\exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$

Find $b'(\theta), a(\phi), b''(\theta)$.

Check if $E[\cdot], V(\cdot)$ matches.

X_1	X_2	$\ldots$	X_d	Y
·	·	$\ldots$	·	·
·	·	$\ldots$	·	·
·	·	$\ldots$	·	·
·	·	$\ldots$	·	·

Y has pmf / pdf of the form

$$f(y_i) = \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi_i)} + c(y_i, \phi_i)\right\}$$

where $E[Y_i] = \mu_i = b'(\theta_i)$.

$$\eta_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di}$$

$$\eta_i = g(\mu_i)$$

$\hookrightarrow$ link fns.

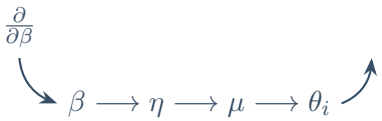
Here the likelihood is given by

$$L = \prod_{i=1}^{n} \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi_i)} + c(y_i, \phi_i)\right\}$$

$$l = \sum_{i=1}^{n}\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi_i)} + c(y_i, \phi_i)\right\}$$

$$l(\beta) = \sum_{i=1}^{n}\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi_i)} + c(y_i, \phi_i)\right\}$$

$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^{n}\frac{\partial}{\partial \beta}\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi_i)} + c(y_i, \phi_i)\right\}$$



$$= \sum_{i=1}^{n} \frac{\partial \eta_i}{\partial \beta} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial}{\partial \theta_i} \{\dots\}$$

$$= \sum_{i=1}^{n} \left(\frac{y_i - b'(\theta_i)}{a(\phi_i)} \right) \frac{1}{b''(\theta_i)} \times \frac{1}{g'(\mu_i)} \times \underline{x}_i$$

$$\eta_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_d X_{di}$$
$$= \underline{\beta}' \underline{x}_i \qquad [\underline{x}_i = (1 \;\; X_{1i} \;\; \dots \;\; X_{di})]$$

$$\mu_i = b'(\theta_i)$$
$$\frac{\partial \mu_i}{\partial \theta_i} = b''(\theta_i)$$
$$\eta_i = g(\mu_i)$$
$$\frac{\partial \eta_i}{\partial \mu_i} = g'(\mu_i)$$
$$\frac{\partial \eta_i}{\partial \beta} = \dots$$

$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^{n} \frac{y_i - \mu_i}{\mathrm{Var}(y_i)} \frac{1}{g'(\mu_i)} \underline{x}_i \qquad (\, l(\beta) = \text{log-likelihood funct.}\,)$$

$\underline{\beta}^{(0)}$

$\underline{\beta}^{(1)} = \underline{\beta}^{(0)} - \left[H(\underline{\beta}^{(0)}) \right]^{-1} S(\underline{\beta}^{0})$

(Newton-Raphson)

$\underline{\beta}^{(1)} = \underline{\beta}^{(0)} + I(\underline{\beta}^{(0)})^{-1} S(\underline{\beta}^{0})$

(Fisher's Scoring)

$$f(x) = 0$$
$$0 = f(x_0) + (x - x_0) f'(x_0)$$
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$\max l(x)$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$E\left[\left(\frac{\partial l}{\partial \theta}\right)^2\right] = E\left[-\frac{\partial^2 l}{\partial \theta^2}\right] \qquad \text{(scalar version)}$$

$$E\left[\frac{\partial l}{\partial \beta}\left(\frac{\partial l}{\partial \beta}\right)^T\right] = I(\beta) \qquad \text{(vector version)}$$

$$I(\beta) = E\left[\left(\sum_{i=1}^n \frac{y_i-\mu_i}{V(y_i)}\frac{1}{g'(\mu_i)}\underline{x}_i\right)\left(\sum_{i=1}^n \frac{y_i-\mu_i}{V(y_i)}\frac{1}{g'(\mu_i)}\underline{x}_i\right)^T\right].$$

Now, $$E\left[\left(\sum_{i=1}^n \underline{z}_i\right)\left(\sum_{i=1}^n \underline{z}_i\right)^T\right] = E\left[(\underline{z}_1+\underline{z}_2+\cdots+\underline{z}_n)(\underline{z}_1+\cdots+\underline{z}_n)^T\right]$$

$$= E\left[\sum_{i=1}^n\sum_{j=1}^n \underline{z}_i\underline{z}_j^T\right]$$

$$= \sum_{i=1}^n\sum_{j=1}^n E\left[\underline{z}_i\underline{z}_j^T\right]$$

$$I(\beta) = E\left[\sum_{i=1}^n \frac{(y_i-\mu_i)^2}{V^2(y_i)}\frac{1}{(g'(\mu_i))^2}x_i x_i^T\right] \qquad \begin{pmatrix}\text{If } i\neq j,\\ E[(y_i-\mu_i)(y_j-\mu_j)]\\ =0\end{pmatrix}$$

$$= \sum_{i=1}^n \frac{\cancel{V(y_i)}}{\cancel{V^2}(y_i)}\frac{1}{(g'(\mu_i))^2}x_i x_i^T$$

$$\therefore \quad S(\underline{\beta}) = \sum_{i=1}^n \frac{y_i-\mu_i}{V(y_i)}\frac{1}{g'(\mu_i)}\underline{x}_i$$

$$I(\underline{\beta}) = \sum_{i=1}^n \frac{1}{V(y_i)}\frac{1}{(g'(\mu_i))^2}\underline{x}_i\underline{x}_i^T$$

$$z_i - \eta_i = (y_i-\mu_i)g'(\mu_i) \quad i=1,\ldots,n$$

$$I(\beta) = X'DX.$$

$$S(\underline{\beta}) = \sum_{i=1}^n \frac{z_i-\eta_i}{V(y_i)}\frac{1}{(g'(\mu_i))^2}\underline{x}_i$$

$$= X'D(\underline{z}-\underline{\eta}), \quad \underline{z} = \begin{pmatrix} z_1\\ \vdots\\ z_n\end{pmatrix},\ \underline{\eta} = \begin{pmatrix}\eta_1\\ \vdots\\ \eta_n\end{pmatrix}$$

Define

$$\underline{X} = \begin{bmatrix}\underline{x}_1^T\\ \underline{x}_2^T\\ \vdots\\ \underline{x}_n^T\end{bmatrix} = \begin{bmatrix}1 & X_{11} & X_{21} & \ldots & X_{d1}\\ 1 & X_{12} & X_{22} & \ldots & X_{d2}\\ \vdots & \vdots & \vdots & & \vdots\\ 1 & X_{1n} & X_{2n} & \ldots & X_{dn}\end{bmatrix}$$

$$D = \begin{bmatrix}\frac{1}{V(y_1)(g'(\mu_1))^2} & & 0\\ & \ddots & \\ 0 & & \frac{1}{V(y_n)(g'(\mu_n))^2}\end{bmatrix}$$

$z_i = \eta_i + (y_i-\mu_i)g'(\mu_i)$ are

$$g(y_i) = z_i = \underbrace{g(\mu_i)}_{\eta_i} + (y_i-\mu_i)g'(\mu_i)$$

$i=1,\ldots,n$

$$\underline{\beta}^{(t+1)} = \underline{\beta}^{(t)} + I(\underline{\beta}^{(t)})^{-1}S(\underline{\beta}^{(t)}) \qquad \text{(update rule)}$$

$$= \underline{\beta}^{(t)} + (X'DX)^{-1}X'D(\underline{z}-\underline{\eta}) \qquad \text{(IRLS Iteration)}$$

$\hookrightarrow$ WLS.

$$\begin{array}{l|l} t=0 & \text{Start from } \underset{\sim}{\beta}^{(t)} \text{ for } t=0 \\ \underset{\sim}{\beta}^{(t)} & \text{compute :} \\ & \hookrightarrow D^{(t)},\ z^{(t)},\ \eta^{(t)} \end{array}$$

Then,

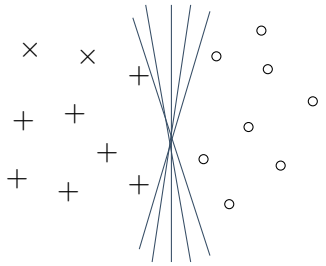
$$\begin{aligned} \underset{\sim}{\beta}^{(t+1)} &= \underset{\sim}{\beta}^{(t)} + (X'D^{(t)}X)^{-1}X'D^{(t)}(\underset{\sim}{z}^{(t)} - \underset{\sim}{\eta}^{(t)}) \\ &= \underset{\sim}{\beta}^{(t)} + (X'D^{(t)}X)^{-1}X'D^{(t)}\underset{\sim}{z}^{(t)} - (X'D^{(t)}X)^{-1}X'D^{(t)}\underset{\sim}{\eta}^{(t)} \end{aligned}$$

$$\begin{cases} \eta_1^{(t)} = \beta_0^{(t)} + \beta_1^{(t)}X_{11} + \cdots + \beta_d^{(t)}X_{d1} = \underset{\sim}{x}_1'\underset{\sim}{\beta}^{(t)} \\ \quad\vdots \\ \eta_n^{(t)} = \underset{\sim}{x}_n'\underset{\sim}{\beta}^{(t)} \end{cases} \qquad \underset{\sim}{\eta}^{(t)} = X\underset{\sim}{\beta}^{(t)}$$

$$\therefore \quad \underset{\sim}{\beta}^{(t+1)} = \cancel{\underset{\sim}{\beta}^{(t)}} + (X'D^{(t)}X)^{-1}X'D^{(t)}\underset{\sim}{z}^{(t)} - \cancel{\underset{\sim}{\beta}^{(t)}}$$

Finally, we get : $\underset{\sim}{\beta}^{(t+1)} = (X'D^{(t)}X)^{-1}X'D^{(t)}\underset{\sim}{z}^{(t)}$.

L-26 Linearly Separable Case & Margin Maximization



We are looking for a line that maximizes the separation b/w these two classes.

	X (measurement)	Y (label)
$\underset{\sim}{X}_1$	$(X_{11} \quad X_{21} \quad \dots \quad X_{d1})$	$y_1 = 1$
$\underset{\sim}{X}_2$		$y_2 = -1 \quad (\times)$
$\vdots$		$\vdots$
$\underset{\sim}{X}_n$		$y_n = 1 \quad (\circ).$

We will begin with assuming that the observations are linearly separable.

Linearly separable case

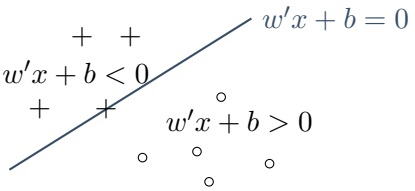
There exists a hyperplane $w'x+b=0$, which perfectly separates the data clouds from the two classes.

If $y_i = 1$ (class $\circ$) then $w'x_i + b \geq 0$.

If $y_i = -1$ (class $\times$) then $w'x_i + b \leq 0$.

$\Rightarrow (w'x_i + b)y_i \geq 0 \quad \forall\, i = 1, 2, \dots, n.$

(This would be one of the conditions)



Our aim is to find $\underset{\sim}{w}$ and b such that

$$(w'x_i + b)y_i \geq M \qquad \forall\, i = 1, \dots, n.$$

for M as large as possible.

We look for maximization of M, s.t. $(w'x_i + b)y_i \geq M \quad \forall\, i = 1, 2, \ldots, n.$

(But this optimization problem is ill-posed / not meaningful.)

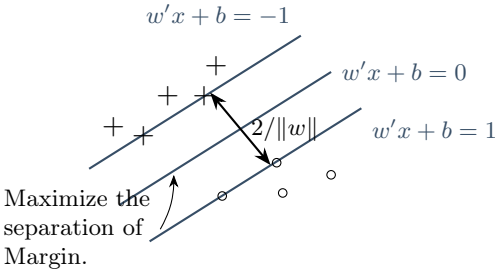
$$\begin{aligned} &\hookrightarrow \quad (w'x_i + b)y_i \geq 10 \quad \text{(say)} \\ &\Leftrightarrow \quad 100 \times (w'x_i + b)y_i \geq 1000 \\ &\Leftrightarrow \quad (w^{*\prime}x_i + b^*)y_i \geq 1000 \qquad \text{(non-identifiable)} \end{aligned}$$

To make it meaningful we can divide by the norm of w.

A meaningful version is maximization of M

$$\text{s.t.} \qquad (w'x_i + b)y_i \geq M\|w\|.$$

Without loss of generality, we can take $M\|w\| = 1$ or $M = \frac{1}{\|w\|}$.



So, we can maximize $\frac{1}{\|w\|}$ s.t. $(w'x_i + b)y_i \geq 1$.

The optimization problem we consider is

Primal

$$\begin{cases} \text{Min} & \frac{1}{2}\|w\|^2 \\ \text{s.t.} & (w'x_i + b)y_i \geq 1 \\ & \forall\, i = 1, 2, \ldots, n \end{cases}$$

$\|\cdot\|^2 \longrightarrow$ convex problem.

$\frac{1}{2}\|\cdot\|^2 \longrightarrow$ derivative would cancel out 2.

$$\text{Lagrangian :} \qquad L(w, b, \underset{\sim}{\lambda}) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \lambda_i\{1 - (w'x_i + b)y_i\}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n \geq 0$.

$$\underset{\sim}{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_n).$$

$$\frac{\partial L}{\partial \underline{w}} = 0 \quad \Rightarrow \quad w - \sum_{i=1}^{n} \lambda_i \underline{x}_i y_i = 0 \quad \Rightarrow \quad w = \sum_{i=1}^{n} \lambda_i \underline{x}_i y_i \quad \longrightarrow ①$$

$$\frac{\partial L}{\partial b} = 0 \quad \Rightarrow \quad -\sum \lambda_i y_i = 0 \quad \Rightarrow \quad \sum_{i=1}^{n} \lambda_i y_i = 0 \quad \longrightarrow ②$$

$$\begin{aligned} g(\lambda) &= \frac{1}{2}\left(\sum \lambda_i \underline{x}_i y_i\right)^T \left(\sum \lambda_i \underline{x}_i y_i\right) + b\cancelto{0}{\sum \lambda_i y_i} - \sum \lambda_i \underline{w}' \underline{x}_i y_i + \sum_{i=1}^{n} \lambda_i \\ &= \frac{1}{2}\left(\sum \lambda_i \underline{x}_i y_i\right)^T \left(\sum \lambda_i \underline{x}_i y_i\right) - \underbrace{\left(\sum \lambda_i \underline{x}_i y_i\right)^T}_{w'} \left(\sum \lambda_i \underline{x}_i y_i\right) + \sum_{i=1}^{n} \lambda_i \\ &= \sum \lambda_i - \frac{1}{2}\sum_{i,j} \lambda_i \lambda_j \underline{x}_i' \underline{x}_j y_i y_j \\ &= \sum \lambda_i - \frac{1}{2}\sum_{i,j} \lambda_i \lambda_j \langle \underline{x}_i, \underline{x}_j \rangle y_i y_j \end{aligned}$$

So, the dual is given by

$$\begin{aligned} \text{Maximize} \quad & \sum_{i=1}^{n} \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j \langle \underline{x}_i, \underline{x}_j \rangle y_i y_j \\ \text{s.t.} \quad & \lambda_1, \ldots, \lambda_n \geq 0 \\ & \sum \lambda_i y_i = 0 \end{aligned}$$

↷ This inner product is needed later in nonlinear classifier.

We can now use a program / solver to get the solution.

Suppose that $\lambda_1^*, \lambda_2^*, \ldots, \lambda_n^*$ is the solution of the dual problem.

$$w = \sum \lambda_i \underline{x}_i y_i \qquad \text{from } ①.$$

$$\text{So,} \qquad w^* = \sum_{i=1}^{n} \lambda_i^* \underline{x}_i y_i$$

Under the KKT condition, we have

$$\lambda_i^* \{1 - (w^{*\prime} x_i + b^*) y_i\} = 0 \quad \forall\, i = 1, \ldots, n.$$

(complementary slackness condition)

One can choose i_0 such that $\lambda_{i_0} > 0$.

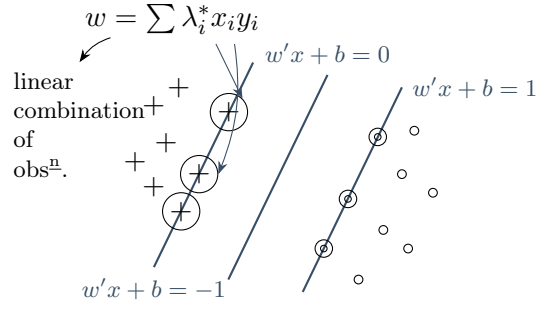
$$\text{For such an } i_0, \qquad 1 - (w^{*\prime} x_{i_0} + b^*) y_i = 0$$
$$(w^{*\prime} x_{i_0} + b^*) y_i = 1.$$

b^* can be obtained from this.

In practice, one computes b^* for different i with $\lambda_i^* > 0$ and takes the average.

After getting λ^*, w^* and b^*, we classify an obs$^{\underline{n}}$ X to the positive (negative, respectively) class if

$$\begin{aligned} & w^{*\prime} x + b^* > 0 \\ \Rightarrow \quad & \sum \lambda_i^* (\underline{x}_i' \underline{x}) y_i + b^* > 0 \\ \Rightarrow \quad & \sum \lambda_i^* \langle \underline{x}_i, \underline{x} \rangle y_i + b^* > 0. \end{aligned}$$



In higher dim. there is more possibility to get data on the boundaries as more flexibility to rotate.

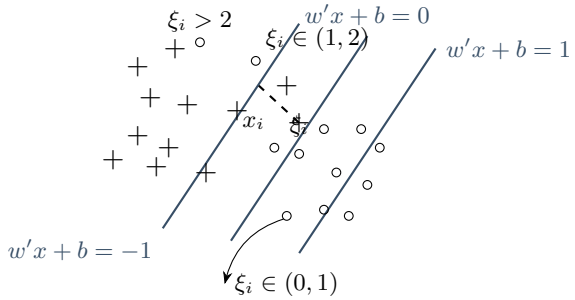
KKT condition

$$\lambda_i \{1 - (w' x_i + b) y_i\} = 0 \quad \forall\, i = 1, \ldots, n.$$

So, it is 0 for those obs. lying on the boundaries.

The observations with $\lambda_i > 0$ are called support vectors.
$\underline{w}$ can be written as a linear comb$^{\underline{n}}$ of these support vectors.

Linear SVM for the non-separable case



Note: $\sum_{i=1}^{n} \xi_i$ serves as an upper bound of the mis-classifications.

$$(w'x_i + b)y_i \geq 1$$
$$\cancel{\text{for all}} \quad i = 1, \ldots, n$$

(Modified)

$$(w'x_i + b)y_i \geq 1 - \xi_i \quad \longleftarrow \text{slack variables.}$$
$$\forall\ i = 1, 2, \ldots, n.$$

In the non-separable case :

Primal problem

$$\begin{cases} \text{We minimize} & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i\ ,\ C > 0 \\ \text{subject to} & (w'x_i + b)y_i \geq 1 - \xi_i \\ & \forall\ i = 1, 2, \ldots, n. \end{cases}$$